

1. Determine the optimum assignment of jobs to workers for the following data.

Job	Worker				
		A	B	C	D
	1	8	6	2	4
	2	6	7	11	10
	3	3	5	7	6
	4	5	10	12	9

2. Processing times (including setup times) and due dates for six jobs waiting to be processed at a work center are given in the following table. Determine the sequence of jobs, the average flow time, average tardiness, and average number of jobs at the work center, for each of these rules:
- FCFS
 - SPT
 - EDD
 - CR

Job	Processing Time (Days)	Due Date
A	2	7
B	8	16
C	4	4
D	10	17
E	5	15
F	12	18

3. A quality inspector took five samples, each with four observations ($n = 4$), of the length of time for glue to dry. The analyst computed the mean of each sample and then computed the grand mean. All values are in minutes. Use this information to obtain three-sigma (i.e., $z = 3$) control limits for means of future times. It is known from previous experience that the standard deviation of the process is .02 minute.

Observation		Sample				
		1	2	3	4	5
	1	12.11	12.15	12.09	12.12	12.09
	2	12.10	12.12	12.09	12.10	12.14
	3	12.11	12.10	12.11	12.08	12.13
	4	12.08	12.11	12.15	12.10	12.12

When the standard deviation is unknown, also calculate the control limits.

Number of Observations in Sample, n	Factor for Chart, A_2	FACTORS FOR R CHARTS	
		Lower Control Limit, D_3	Upper Control Limit, D_4
2	1.88	0	3.27
3	1.02	0	2.57
4	0.73	0	2.28
5	0.58	0	2.11
6	0.48	0	2.00
7	0.42	0.08	1.92
8	0.37	0.14	1.86
9	0.34	0.18	1.82
10	0.31	0.22	1.78
11	0.29	0.26	1.74
12	0.27	0.28	1.72
13	0.25	0.31	1.69
14	0.24	0.33	1.67
15	0.22	0.35	1.65
16	0.21	0.36	1.64
17	0.20	0.38	1.62
18	0.19	0.39	1.61
19	0.19	0.40	1.60
20	0.18	0.41	1.59

Source: Adapted from Eugene Grant and Richard Leavenworth, *Statistical Quality Control*, 5th ed. Copyright © 1980 McGraw-Hill Companies, Inc. Used with permission.